Technical Supplement: Blocked Response Surface Modeling and Multi-Objective XGBoost Optimization

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Purpose of this supplement. This document preserves the complete mathematical derivations, full statistical tables, diagnostic results, timing protocols, provenance explanations, and extended benchmark interpretation that support the shorter reader-focused main report. It reports the same frozen empirical evidence and should be read alongside `report.pdf`.

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Abstract

Hyperparameter optimization in gradient boosted decision trees is commonly conducted using adaptive black-box search methods, which offer strong point optimization performance but limited parametric attribution of factor effects and data-split nuisance variation. We evaluate a classical Design of Experiments (DOE) and Response Surface Methodology (RSM) framework based on Douglas C. Montgomery's methodology to study and optimize an XGBoost regressor across four hyperparameters on the California Housing dataset (N = 20,640) under stochastic nuisance blocking. A pre-planned, globally randomized 140-run Face-Centered Central Composite Design (FCCD) across five deterministic seed blocks isolates data-partition variance ($\sigma_{\text{block}} \approx 0.0077$ RMSE, ICC = 0.4069 of unexplained residual variance) and is analyzed in two hierarchical stages: factorial-plus-center screening ($N_1 = 100$) and full second-order augmentation ($N_{\text{CCD}} = 140$, where 30 of 40 axial runs occur at `run_order` ≤ 100). Screening detects strong response curvature ($\bar{y}_F - \bar{y}_C = +0.1343$ RMSE, +26.93%; $F = 105432.31$ against within-block center subsampling error, F = 20,235.41 against 79-df block-adjusted pooled error, and F = 7,342.69 across 5 paired blocks), while second-order lack-of-fit decomposition ($F = 62.21$) shows that quadratic polynomial surrogates serve as local guidance maps rather than globally exact estimators of boosting loss. Canonical analysis reveals a stationary/rising ridge system, and Derringer-Suich desirability optimization identifies a compromise coordinate ($\mathbf{x}^*_{\text{MO}}$, depth 4) confirmed across $m = 10$ fresh seeds, whereas the single-objective boundary candidate ($\mathbf{x}^*_{\text{SO}}$, depth 7) exhibits $++\,0.0217$ RMSE polynomial optimism bias. In a prospective 20-replicate benchmark campaign (17,077 model fits, 1,640,810 timed inferences; 122 frozen selection records across 95 distinct configurations evaluated on 20 fresh seeds on the 4,128-row holdout test set), Single-Objective TPE achieves the lowest observed test RMSE (0.46691 ± 0.00116) by concentrating all 140 adaptive trials in deeper trees ($d \in \{7, 8, 9\}$) at higher latency ($188.93 \pm 12.07\,\mu$s), whereas Repeated DOE Single-Objective yields deterministic boundary selection (0.46859, conditional retraining $\sigma_{\text{eval}} = 0.00304$) with 20.2% lower latency ($150.74 \pm 0.72\,\mu$s). In latency-constrained multi-objective optimization ($\leq 145\,\mu$s), Repeated DOE and Multi-Objective TPE show no statistically significant difference in holdout test RMSE (0.48997 vs. 0.49255, primary Welch $p = 0.3826$) at comparable latency (122.3 vs. 121.5 μs) and 100% (20/20) feasibility, while contributing jointly to a 12-point holdout non-dominated frontier (seed-bootstrap HV $= 17.6923 \pm 0.0564$).

Keywords: Design of Experiments (DOE), Response Surface Methodology (RSM), Central Composite Design (CCD), Gradient Boosted Trees, XGBoost, Multi-Objective Optimization, Randomized Complete Block Design (RCBD), Canonical Analysis, Model Adequacy.

1 Introduction and Theoretical Framework

Gradient Boosted Decision Trees (GBDTs), such as XGBoost [2], achieve competitive predictive accuracy on tabular data. However, tuning GBDT hyperparameters involves non-linear interactions, non-convex loss surfaces, and operational trade-offs between predictive accuracy and single-sample inference latency.

Standard practice relies on heuristic and adaptive search algorithms, including uniform random search [3] and Bayesian optimization methods such as Tree-structured Parzen Estimators (TPE) [4, 5] or Gaussian Process regression. In practical workflows, these black-box optimizers present three structural trade-offs relative to parametric response surface methods:

(i) **Absence of Parametric Attribution:** Black-box optimizers treat the objective function as an opaque evaluation oracle. They do not decompose variance across main effects, two-factor interactions, and quadratic terms, nor do they provide a closed-form polynomial surrogate for canonical sensitivity analysis.

(ii) **Confounding of Stochastic Nuisance:** Variation arising from random data partitioning and stochastic subsampling is confounded with hyperparameter sensitivity unless structured blocking or repeated multi-split evaluation is introduced.

(iii) **Lack of Curvature and Adequacy Tests:** Heuristic search strategies do not provide formal ANOVA hypothesis tests to determine whether an operational region is planar, locally quadratic, or structurally misfitted by low-order polynomials.

To evaluate these trade-offs against adaptive search under matched evaluation budgets, this study implements a pre-planned, two-stage hierarchical Design of Experiments (DOE) and Response Surface Methodology (RSM) pipeline based on Chapters 4, 5, 6, 7, 10, 11, and 13 of Douglas C. Montgomery's *Design and Analysis of Experiments* [1], following response surface methodology developed by Box and Wilson [7] and extended by Myers et al. [8]. By combining randomized complete block screening, multi-denominator curvature testing, Face-Centered Central Composite Design (FCCD) modeling, spectral canonical analysis, and Derringer-Suich desirability functions [6], we examine both the variance-attribution strengths and the boundary-allocation and polynomial-misfit limitations of classical RSM on GBDT hyperparameter surfaces.

All metrics, sums of squares, test statistics, and diagnostics reported in this paper are derived from model fits on the California Housing benchmark ($N = 20,640$). An external holdout test set of 4,128 observations (20%) was set aside from the 16,512-observation development pool, which each seed block partitioned into 12,384 training (75%) and 4,128 validation (25%) samples. Whereas the historical $v1.0.0$ experiments recorded test results during development, the revised pipeline prevented access to those test labels during all surrogate fitting, curvature testing, lack-of-fit analysis, and desirability optimization until winning configurations were frozen in `finalized_selections.json`. These safeguards enforce programmatic holdout isolation during the revised search, although they cannot retroactively undo earlier exposure of the same dataset.

2 Experimental Architecture and Mathematical Formulation

2.1 Operational Factors and Standard Coding

We define four hyperparameter factors (three continuous, one integer depth) controlling structural complexity and regularization for an XGBoost regressor with $n_{\text{estimators}} = 100$:

- **Factor A (x_1):** Learning rate ($\eta \in [0.01, 0.30]$), coded on a logarithmic scale: $\xi_1 = \ln(\eta) \in [\ln(0.01), \ln(0.30)] = [-4.6052, -1.2040]$.
- **Factor B (x_2):** Maximum tree depth (depth $\in [3, 9]$), controlling structural model complexity.
- **Factor C (x_3):** Subsample fraction (subsample $\in [0.50, 1.00]$), introducing stochastic row subsampling.
- **Factor D (x_4):** L_2 leaf regularization ($\lambda \in [0.10, 10.00]$), coded on a logarithmic scale: $\xi_4 = \ln(\lambda) \in [\ln(0.10), \ln(10.00)] = [-2.3026, 2.3026]$.

Each natural variable ξ_i is mapped to dimensionless coded space $x_i \in [-1, +1]$ via:

$$x_i = \frac{\xi_i - \frac{\xi_{i,\max} + \xi_{i,\min}}{2}}{\frac{\xi_{i,\max} - \xi_{i,\min}}{2}}, \quad i \in \{1, 2, 3, 4\} \tag{1}$$

The operational domain forms a 4-dimensional hypercube $\mathcal{D} = [-1, +1]^4$. Table 1 summarizes factor coding.

Factor	Natural Variable	Transform	Low -1	Center 0	High $+1$
Factor A (x_1)	Learning Rate (η)	$\ln(\eta)$	0.0100	0.0548	0.3000
Factor B (x_2)	Max Tree Depth	Linear	3	6	9
Factor C (x_3)	Subsample Fraction	Linear	0.5000	0.7500	1.0000
Factor D (x_4)	L_2 Reg (λ)	$\ln(\lambda)$	0.1000	1.0000	10.0000

Table 1: Hyperparameter Factors, Operational Domains, and Design Levels

2.2 Randomized Complete Block Design (RCBD), Study Chronology, and Budget Allocation

Stochastic variation in data partitioning and subsampling introduces non-trivial variance across evaluation trials. We structure the experimental protocol as a Randomized Complete Block Design (RCBD) across $b = 5$ distinct seed blocks: $\mathcal{S} = \{42, 101, 202, 303, 404\}$. Table 2 summarizes the assignment of split seeds and model seeds across phases.

Phase / Design Point	Point IDs	Block	Split Seed	Model Seed	Reps	Isolated Variance
Phase 1: Corner Runs	$1 \dots 16$	$b \in \{1..5\}$	S_b	S_b	1	Factorial + Block
Phase 1: Center Points	17	$b \in \{1..5\}$	S_b	$S_b + 1000r$	4	Subsampling Error (15 df)
Phase 2: Axial Runs	$18 \dots 25$	$b \in \{1..5\}$	S_b	S_b	1	Quadratic + Block

Table 2: Structure of Blocks, Split Seeds, and Model Seeds Across Experimental Phases

Specifically, each block b defines a distinct 75/25 train/validation partition of the 16,512 development observations via split seed S_b. For corner runs ($1 \dots 16$) and axial runs ($18 \dots 25$), the internal XGBoost subsampling seed is also set to S_b, meaning that the block factor captures the joint nuisance impact of data-split selection and stochastic subsampling realization. For the four center point replicates within block b, the train/validation split is held fixed at S_b while the model subsampling seed is varied via $S_b + r \times 1000$ ($r \in \{0, 1, 2, 3\}$), isolating within-block stochastic subsampling pure error (15 degrees of freedom) from block-to-block variation (4 degrees of freedom).

Actual Study Chronology vs. Two-Stage Hierarchical Analysis: In the execution script (`pipeline.py`), all $28 \times 5 = 140$ runs of the Face-Centered Central Composite Design (16 factorial corners, 4 center replicates, and 8 axial points per block) were generated upfront and globally randomized into a single execution schedule (`run_order` $\in \{1, \dots, 140\}$). In this randomized schedule, exactly 30 of the 40 `Phase2_Axial` runs were executed at `run_order` $\leq$ 100, interleaved with `Phase1_Factorial` runs. Thus, Phase 1 ($N_1 = 100$) and Phase 2 ($N_{\text{CCD}} = 140$) denote a two-stage hierarchical statistical decomposition of a pre-planned 140-run FCCD rather than an adaptive sequential design in which axial runs were conditioned on observing Phase 1 outcomes.

Budget Allocation Trade-Off (25 Replicated Coordinates vs. 140 Distinct Configurations): Under the matched 140-evaluation budget per search campaign, the blocked FCCD evaluates 25 distinct hyperparameter coordinates replicated across 5 seed blocks (125 coordinate-

block evaluations plus 15 extra center runs = 140 fits). Because a 2^4 factorial plus face-centered axial design places 8 corners and 1 axial point per block at the lower learning-rate boundary $x_1 = -1.0$ ($\eta = 0.01$), exactly 45/140 runs (32.14%) evaluate $\eta = 0.01$, where 100-tree ensembles underfit (mean validation RMSE 0.7545 vs. 0.5055 for $x_1 > -1.0$). By contrast, adaptive TPE and Random Search allocate all 140 evaluations to distinct hyperparameter configurations on a single validation split without cross-block replication, trading variance-component identifiability and multi-split blocking for denser interior coordinate exploration.

The statistical model for response y_{uj} at design point u in block j is:

$$y_{uj} = \mu + \tau_u + \beta_j + \epsilon_{uj}, \quad j \in \{1, \ldots, 5\} \tag{2}$$

where τ_u represents the treatment effect of design coordinate $\mathbf{x}_u$, β_j represents the block effect, and ϵ_{uj} is experimental error.

To quantify the magnitude of variance absorbed by blocking, we compute the Intraclass Correlation Coefficient (ICC) under both ANOVA and REML formulations:

$$\text{ICC} = \frac{\sigma^2_{\text{block}}}{\sigma^2_{\text{block}} + \sigma^2_{\epsilon}} \tag{3}$$

For validation RMSE, the ANOVA estimator yields ICC = 0.4069 ($F = 20.21, p < 0.0001$), which matches the restricted maximum likelihood estimate ($\text{ICC}_{\text{REML}} = 0.4069$). Importantly, this 40.7% represents a share of *unexplained residual variance* rather than total experimental variation (where primary factors account for $> 90\%$ of total sum of squares). In absolute terms, the estimated block standard deviation is $\sigma_{\text{block}} = \sqrt{\sigma^2_{\text{block}}} \approx 0.0077$ RMSE. This nuisance variation is comparable in magnitude to the performance differences separating competitive hyperparameter configurations (often 0.005–0.015 RMSE), which makes blocking necessary when comparing close configurations.

While the block effect is non-significant in Phase 1 ($F = 0.33, p = 0.8541$), it is significant in the Phase 2 second-order model ($F = 20.21, p < 10^{-11}$). This difference reflects variance shielding: in Phase 1, unmodeled quadratic curvature pooled into the residual denominator ($\text{MS}_E = 0.003485$), inflating error variance and attenuating the block F-ratio. Once second-order terms accounted for curvature in Phase 2 (MS_E reduced to 0.000086), seed partition variance became detectable. For inference latency, the block effect is negligible ($\text{ICC} = 0.0000, F = 0.75, p = 0.5591$), as execution speed is governed by hardware architecture and tree topology rather than data splits.

2.3 Dual Responses and Latency Measurement Protocols

For every trial, two responses are measured:

1. **Response Y_1 (Development Validation RMSE):** Root mean squared error evaluated on the 25% validation partition ($N_{\text{val}} = 4,128$) of the development pool. Unlike the historical $v1.0.0$ pipeline, which recorded test metrics during development, the revised pipeline prevented access to the holdout 20% test partition ($N_{\text{test}} = 4,128$) until candidate configurations were frozen for final generalization testing.

2. **Response Y_2 (Single-Sample Inference Latency):** Execution time in microseconds per single sample (μs/sample) on CPU. We emphasize that this represents single-sample point evaluation rather than batched throughput, and the quadratic depth term is an empirical response-surface fit across sampled depths $d \in \{3, 6, 9\}$.

Latency measurements across the study follow three explicitly distinguished protocol tiers:

(i) **Pinned CPU Affinity and Single-Threaded Execution:** Across all historical and prospective scripts, the execution process is pinned to CPU Core 0 via Windows API `SetProcessAffinityMask(GetCurrentProcess(), 1)` with single-threaded XGBoost inference (`n_jobs=1`) timed via `time.perf_counter_ns`.

(ii) **Historical DOE, Confirmation, and Baseline Protocol (`results/runs.csv`, `confirmation_runs*.csv`, `benchmark.csv`):** Each model fit executes 100 untimed warmup calls (50 per prediction interface) followed by 1,000 timed single-sample calls per interface.

(iii) **Prospective Online Search vs. Holdout Verification Protocols (`full_run_001`):** During prospective search (`optimizer_trials.csv` and `doe_selection_runs.csv`), candidate models are timed using 30 un-warmed single-row `predict` calls per fit (`online_30call_single_row_cpu0`) to bound wall-clock cost across 14,000 search fits. During final holdout verification (`final_evaluations.csv`), each frozen configuration is timed across 2 independent sessions of 25 warmup calls per interface (50 total) and 500 timed single-row calls per interface per session (1,000 timed calls/seed; between-session SD = 1.28 μs).

(iv) **Interface Overhead (`Booster.predict(DMatrix)` vs. `XGBRegressor.inplace_predict`):** Standard `predict` constructs a `DMatrix` wrapper and validates 2D array metadata on each single-row call, whereas `inplace_predict` operates directly on the 2D NumPy buffer. The measured per-sample overhead of `predict` minus `inplace_predict` is $40.88 \pm 11.56\,\mu$s (range 5.75–89.33 μs) across the 140 historical DOE runs, $38.18 \pm 5.24\,\mu$s ($\mathbf{x}^*_{\text{MO}}$) and $36.85 \pm 7.75\,\mu$s ($\mathbf{x}^*_{\text{SO}}$) in historical confirmation, $32.07 \pm 1.16\,\mu$s (range 30.89–33.96 μs) in the historical benchmark, and $32.19 \pm 1.46\,\mu$s (range 27.20–37.43 μs) in the 50-configuration Revision-v2 interface audit (`latency_interface_overhead.json`).

3 Phase 1: Factorial Screening and Curvature Assessment

Phase 1 analyzes the 2^4 full factorial subset consisting of $n_F = 16$ corner runs, augmented with $n_C = 4$ center points per block ($x_1 = x_2 = x_3 = x_4 = 0$). Evaluated across all 5 seed blocks, this screening stage comprises $N_1 = 100$ model training runs.

We fit a first-order response surface model with all two-factor interaction terms:

$$Y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{b=2}^{5} \gamma_b Z_b + \epsilon \tag{4}$$

where Z_b are dummy indicators for blocks 2 through 5. Table 3 details the Type III Sum of Squares ANOVA for Response Y_1.

Source of Variation	SS	DF	MS	F-Stat	p value	Partial η^2
Intercept	7.309127	1	7.309127	2097.15	$< 10^{-15}$	0.9610
Block Effect C(block)	0.004662	4	0.001166	0.33	0.8541	0.0155
Factor A: x_1 ($\ln(\eta)$)	1.245625	1	1.245625	357.40	$< 10^{-15}$	0.8079
Factor B: x_2 (Depth)	0.082155	1	0.082155	23.57	0.0000	0.2171
Factor C: x_3 (Subsample)	0.002406	1	0.002406	0.69	0.4084	0.0081
Factor D: x_4 ($\ln(\lambda)$)	0.000089	1	0.000089	0.03	0.8731	0.0003
Interaction $x_1 \cdot x_2$	0.091725	1	0.091725	26.32	0.0000	0.2364
Interaction $x_1 \cdot x_3$	0.001878	1	0.001878	0.54	0.4649	0.0063
Interaction $x_1 \cdot x_4$	0.007452	1	0.007452	2.14	0.1474	0.0245
Interaction $x_2 \cdot x_3$	0.001106	1	0.001106	0.32	0.5747	0.0037
Interaction $x_2 \cdot x_4$	0.000092	1	0.000092	0.03	0.8714	0.0003
Interaction $x_3 \cdot x_4$	0.000081	1	0.000081	0.02	0.8790	0.0003
Residual Error	0.296247	85	0.003485	—	—	—

Table 3: Phase 1 ANOVA Table (Type III Sum of Squares) for Validation RMSE (Y_1)

Factor A ($\ln(\eta)$) dominates predictive performance ($F = 357.40$, partial $\eta^2 = 0.8079$), followed by Factor B (tree depth, $F = 23.57$) and the interaction $x_1 \cdot x_2$ ($F = 26.32, p < 10^{-6}$).

3.1 Formal Curvature Hypothesis Test and Error-Denominator Sensitivity

Following Montgomery (Chapter 11), overall quadratic curvature $H_0 : \sum_{i=1}^{4} \beta_{ii} = 0$ is tested via the single-degree-of-freedom contrast between the factorial corner mean ($\bar{y}_F$) and the center-point mean ($\bar{y}_C$):

$$\mathrm{SS}_{\mathrm{Curvature}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C} \tag{5}$$

Across the 100 runs evaluated in Phase 1:

$$n_F = 80, \quad \bar{y}_F = 0.6333, \quad n_C = 20, \quad \bar{y}_C = 0.4989, \quad \bar{y}_F - \bar{y}_C = +0.1343 \ (+26.93\%) \tag{6}$$

yielding $\mathrm{SS}_{\mathrm{Curvature}} = 0.288786$. When tested against within-block center-point pure error ($\mathrm{df} = 15, \mathrm{MS}_{\mathrm{PE,\ Center}} = 2.74e-06$), the resulting F-ratio is:

$$F_{\mathrm{Curvature}} = \frac{\mathrm{MS}_{\mathrm{Curvature}}}{\mathrm{MS}_{\mathrm{PE,\ Center}}} = 105432.31 \quad (p < 10^{-15}) \tag{7}$$

Why is $F_{\mathrm{Curvature}} = 105432.31$ of order 10^5? Two structural features of the design explain this magnitude: (i) the numerator ($\mathrm{SS}_{\mathrm{Curvature}} = 0.288786$) captures a macro-scale hyperparameter contrast across the full cube where half of all corner runs lie at $\eta = 0.01$ and severely underfit ($\bar{y}_F = 0.6333$ vs. $\bar{y}_C = 0.4989$), and (ii) the denominator ($\mathrm{MS}_{\mathrm{PE,\ Center}} = 2.74e-06$, SD = 0.0017 RMSE) isolates micro-scale stochastic row-subsampling variability ($s = 0.75$) on a *fixed* train/validation partition S_b, excluding split-by-treatment interaction and higher-order factorial terms.

To ensure that curvature inference does not rely on this narrow subsampling denominator, we evaluate $\mathrm{SS}_{\mathrm{Curvature}} = 0.288786$ across six broader error specifications:

(i) **Block-Adjusted Pooled Coordinate Error (df = 79):** Pooling within-block center pure error (15 df) with the Phase 1 Treatment × Block interaction (64 df) yields MS = 1.43×10^{-5} and $F = 20{,}235.41$ ($p < 10^{-15}$).

(ii) **Phase 1 Treatment × Block Interaction (df = 64):** Testing directly against coordinate-by-split interaction across the 17 × 5 cells (MS = 1.70×10^{-5}) yields $F = 17{,}013.24$ ($p < 10^{-15}$).

(iii) **Block-Paired Contrast Across the 5 Seed Blocks (df = 4):** Computing $d_b = \bar{y}_{F,b} - \bar{y}_{C,b}$ within each block $b \in \{1, \ldots, 5\}$ treats the block as the independent experimental unit, yielding $\bar{d} = +0.1343$ RMSE (SE = 0.0016, 95% CI $[+0.1300, +0.1387]$), paired $t = 85.69$, and $F = t^2 = 7{,}342.69$ ($p = 1.11 \times 10^{-7}$).

(iv) **Unadjusted Coordinate Pure Error (df = 83):** Pooling within-coordinate variance across all 17 Phase 1 coordinates without removing block main effects ($\text{MS}_{\text{PE, All}} = 6.98 \times 10^{-5}$) yields $F = 4{,}140.06 \approx 4{,}140$ ($p < 10^{-15}$).

(v) **First-Order + 2FI + Curvature Residual (df = 84):** Testing against the residual of the 16-parameter block + main + 2FI + curvature model (MS = 8.88×10^{-5}, which pools 3FI+4FI interactions, Treatment × Block, and center pure error) yields $F = 3{,}251.19$ ($p < 10^{-15}$).

(vi) **Unadjusted First-Order + 2FI Residual (df = 85):** Even against the unadjusted Phase 1 residual mean square before separating curvature ($\text{MS}_E = 0.003485$), the ratio is $F = 82.86$ ($p = 3.34 \times 10^{-14}$).

Under every error specification, planarity is rejected by a wide margin. Crucially, rejecting $H_0 : \sum \beta_{ii} = 0$ establishes the *presence of strong non-planarity* across $[-1, +1]^4$; it does not establish that a second-order polynomial is an adequate global functional form. Surrogate predictive adequacy is tested separately via Phase 2 lack-of-fit decomposition (Section 4.2) and Phase 5 confirmation trials (Section 8.1).

4 Phase 2: Second-Order Response Surface Modeling and Adequacy

4.1 Face-Centered Central Composite Design (FCCD)

Incorporating the $2k = 8$ face-centered axial coordinates ($\alpha = 1.0$) across the 5 seed blocks (40 axial runs, cumulative $N_{\text{CCD}} = 140$ runs across 25 distinct coordinates) enables estimation of the full second-order polynomial response surface:

$$\begin{aligned} Y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_{ii} x_i^2 \\ + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{b=2}^{5} \gamma_b Z_b + \epsilon \end{aligned} \tag{8}$$

Table 4 reports the second-order ANOVA for Validation RMSE ($R^2 = 0.9951$, Adj $R^2 = 0.9944$). Because residual variance varies across the factor domain, both OLS and HC3 heteroscedasticity-consistent standard errors and p-values are reported.

Source of Variation	**SS**	**DF**	**MS**	**F-Stat**	**OLS p**	**HC3 SE**	**HC3 p**
Intercept	4.718852	1	4.718852	54749.42	$< 10^{-15}$	0.0021	$< 10^{-15}$
Block Effect C(block)	0.006966	4	0.001742	20.21	< 0.0001	—	—
Factor A: x_1 ($\ln(\eta)$)	1.388673	1	1.388673	16111.76	$< 10^{-15}$	0.0012	$< 10^{-15}$
Factor B: x_2 (Depth)	0.101706	1	0.101706	1180.02	$< 10^{-15}$	0.0013	$< 10^{-15}$
Factor C: x_3 (Subsample)	0.002013	1	0.002013	23.36	< 0.0001	0.0012	0.0001
Factor D: x_4 ($\ln(\lambda)$)	0.000044	1	0.000044	0.52	0.4743	0.0012	0.5483
Quadratic x_1^2	0.146265	1	0.146265	1697.01	$< 10^{-15}$	0.0023	$< 10^{-15}$
Quadratic x_2^2	0.008779	1	0.008779	101.85	$< 10^{-15}$	0.0035	< 0.0001
Quadratic x_3^2	0.000005	1	0.000005	0.05	0.8188	0.0023	0.8005
Quadratic x_4^2	0.000006	1	0.000006	0.08	0.7844	0.0020	0.7197
Interaction $x_1 \cdot x_2$	0.091725	1	0.091725	1064.22	$< 10^{-15}$	0.0013	$< 10^{-15}$
Interaction $x_1 \cdot x_3$	0.001878	1	0.001878	21.79	< 0.0001	0.0013	0.0003
Interaction $x_1 \cdot x_4$	0.007452	1	0.007452	86.46	$< 10^{-15}$	0.0013	< 0.0001
Interaction $x_2 \cdot x_3$	0.001106	1	0.001106	12.83	0.0005	0.0013	0.0050
Interaction $x_2 \cdot x_4$	0.000092	1	0.000092	1.07	0.3040	0.0013	0.4118
Interaction $x_3 \cdot x_4$	0.000081	1	0.000081	0.94	0.3334	0.0013	0.4400
Residual Error	0.010429	121	0.000086	—	—	—	—

Table 4: Full Second-Order ANOVA Table for Validation RMSE (Y_1) with HC3 Robust Standard Errors

The ANOVA confirms substantial quadratic curvature along x_1^2 ($F = 1{,}697.01, p < 10^{-15}$; HC3 $p < 10^{-15}$) and x_2^2 ($F = 101.85, p < 10^{-15}$; HC3 $p < 0.0001$). Under HC3 covariance estimation, structural effects $x_1, x_2, x_1^2, x_2^2, x_1x_2$, and x_1x_4 remain significant ($p < 0.0001$). Table 5 provides the corresponding ANOVA for Inference Latency (Y_2).

Source of Variation	SS	DF	MS	F-Stat	p value
Intercept	555450.04	1	555450.04	1903.08	$< 10^{-15}$
Block Effect C(block)	876.94	4	219.24	0.75	0.5591
Factor A: x_1 ($\ln(\eta)$)	1265.46	1	1265.46	4.34	0.0394
Factor B: x_2 (Depth)	236841.33	1	236841.33	811.46	$< 10^{-15}$
Factor C: x_3 (Subsample)	179.04	1	179.04	0.61	0.4350
Factor D: x_4 ($\ln(\lambda)$)	483.26	1	483.26	1.66	0.2006
Quadratic x_1^2	1576.98	1	1576.98	5.40	0.0218
Quadratic x_2^2	11634.52	1	11634.52	39.86	< 0.0001
Quadratic x_3^2	8.35	1	8.35	0.03	0.8660
Quadratic x_4^2	116.71	1	116.71	0.40	0.5284
Interaction $x_1 \cdot x_2$	662.90	1	662.90	2.27	0.1344
Interaction $x_1 \cdot x_3$	14.00	1	14.00	0.05	0.8270
Interaction $x_1 \cdot x_4$	829.24	1	829.24	2.84	0.0945
Interaction $x_2 \cdot x_3$	813.68	1	813.68	2.79	0.0976
Interaction $x_2 \cdot x_4$	47.57	1	47.57	0.16	0.6871
Interaction $x_3 \cdot x_4$	52.53	1	52.53	0.18	0.6721
Residual Error	35316.19	121	291.87	—	—

Table 5: Full Second-Order ANOVA Table for Inference Latency (Y_2, μs/sample)

4.2 Lack-of-Fit Decomposition and Structural Model Adequacy

In response surface methodology [1], evaluating model adequacy requires decomposing the model residual sum of squares into pure experimental error and structural lack of fit. In our blocked central composite design ($N = 140$ runs across 5 blocks), non-center design coordinates are observed once per block. Genuine within-coordinate and within-block replication exists exclusively at the center points ($x_1 = x_2 = x_3 = x_4 = 0$), where 4 model-seed replicates within each of the 5 blocks provide exactly $\mathrm{df}_{\mathrm{PE,\ center}} = 5 \times (4 - 1) = 15$ degrees of freedom of genuine within-cell pure replicate error ($\mathrm{SS}_{\mathrm{PE,\ center}} = 0.000041$, $\mathrm{MS}_{\mathrm{PE,\ center}} = 2.74e-06$).

Fitting a saturated additive cell model containing all $m = 25$ unique design coordinates and 4 block parameters (29 parameters) leaves $140-29 = 111$ degrees of freedom with SS = 0.001579 (MS = $1.42e-05$). Rather than representing uniform replicate noise, these 111 df decompose into:

(i) Genuine center-point pure replicate error (15 df, SS = 0.000041);

(ii) Treatment × Block interaction / split-specific treatment response ($111 - 15 = 96$ df, $\mathrm{SS} = 0.001538, \mathrm{MS} = 1.60e-05$).

Evaluating the interaction against center pure error yields $F = 5.85$ ($p = 0.0002$), establishing statistically significant split-by-hyperparameter interaction: different train/validation data splits exhibit subtle variations in optimal response curvature.

The remaining $121-111 = 10$ df represent structural lack of fit ($\mathrm{SS}_{\mathrm{LoF}} = 0.008850, \mathrm{MS}_{\mathrm{LoF}} = 0.000885$). Table 6 details this complete decomposition for both responses.

Source of Variation / Model	SS	DF	MS	F	p value	Reference / RMS Misfit
Decomposition of Second-Order Residual (Y_1: Validation RMSE)						
Structural Lack of Fit	0.008850	10	0.000885	323.10	$< 10^{-15}$	vs. Center PE
Treatment × Block Interaction	0.001538	96	1.60×10^{-5}	5.85	0.0002	vs. Center PE
Genuine Center Pure Error	0.000041	15	2.74×10^{-6}	—	—	Base Replicate
Total Model Residual	0.010429	121	8.62×10^{-5}	—	—	RMS = 0.0093
Saturated Additive Baseline	0.001579	111	1.42×10^{-5}	62.21*	$< 10^{-15}$	(*LoF vs. Additive)
Restricted Domain (Y_1: $x_1 \geq -0.5$, $\eta \geq 0.033$, $N = 95$)						
Structural Lack of Fit	0.000324	2	0.000162	10.11	0.0001	RMS = 0.0127
Additive Baseline Residual	0.001203	75	1.60×10^{-5}	—	—	Saturated Base
Total Restricted Residual	0.001528	77	1.98×10^{-5}	—	—	RMS = 0.0045
Latency Residual Decomposition (Y_2: Latency, μs)						
Structural Lack of Fit	3322.37	10	332.24	1.20	0.3625	vs. Center PE (RMS = 18.23)
Treatment × Block Interaction	27844.93	96	290.05	1.05	0.4906	vs. Center PE
Genuine Center Pure Error	4148.89	15	276.59	—	—	Base Replicate
Total Model Residual	35316.19	121	291.87	1.15*	0.3305	(*LoF vs. Additive)

Table 6: Residual Variance Decomposition (121 df), Lack-of-Fit Tests, and Domain Restriction Sensitivity

For Validation RMSE (Y_1) on the full operational domain:

- Tested against center pure error (15 df), lack of fit is statistically significant ($F = 323.10, p < 10^{-15}$).
- Tested against the saturated additive baseline (111 df, pooling interaction and replicate noise), lack of fit is also significant ($F = 62.21, p = 7.78e - 41$).

Under either reference baseline, the root-mean-square misfit per degree of freedom is $\sqrt{\text{MS}_{\text{LoF}}} = 0.0297$, which is approximately 7.8× the additive residual standard deviation ($\sqrt{\text{MS}_{\text{PE}}} = 0.0038$) and 17.5× the center pure error standard deviation ($\sqrt{\text{MS}_{\text{PE, center}}} = 0.0017$). By contrast, the unweighted root-mean-square cell prediction error across the 25 unique design points is $\sqrt{\frac{1}{25}\sum_{i=1}^{25}(\bar{y}_i - \hat{y}_i)^2} \approx 0.0084$, while the observation-weighted root-mean-square misfit across all 140 design evaluations is $\sqrt{\text{SS}_{\text{LoF}}/140} \approx 0.0080$ (down-weighting non-center cells relative to the replicated center).

To inspect where this lack of fit concentrates across the hypercube, per-design-point residual contributions $n_i(\bar{y}_i - \hat{y}_i)^2$ reveal that the single largest lack-of-fit discrepancy occurs at corner $(1, 1, -1, -1)$ ($x_1 = 1, x_2 = 1, x_3 = -1, x_4 = -1$, contribution ≈ 0.00153), where high learning rate and depth combined with low subsampling and regularization challenge second-order surface adequacy. Substantial discrepancies also concentrate at low learning rates ($x_1 = -1, \eta = 0.01$), notably at $(-1, -1, 1, 1)$ (0.00097) and $(-1, 1, -1, 1)$ (0.00070), reflecting underfitting when 100 boosting iterations are paired with low step sizes. When refitting on the restricted sub-domain $x_1 \geq -0.5$ ($\eta \geq 0.033$), the sum of squares of lack of fit drops by 96.3% (from 0.008850 to 0.000324), and the RMS misfit declines by 57% to 0.0127. However, we emphasize that dropping $x_1 = -1$ leaves only two factor levels ($x_1 \in \{0, 1\}$). Because two levels induce the identity $x_1^2 \equiv x_1$, separate linear and quadratic coefficients along Factor A are unidentifiable from those levels alone, reducing df_{LoF} from 10 to 2. Consequently, this restricted test cannot assess quadratic curvature across $x_1 \in [0, 1]$ where all candidate optima reside. Even within this restricted region, lack of fit remains statistically significant ($F = 10.11, p = 0.0001$), with RMS

misfit (0.0127) remaining roughly 3× the additive residual standard deviation (0.0040). This confirms that while low-η underfitting drives the largest discrepancies, a second-order polynomial remains an approximation with non-negligible structural bias across gradient boosting loss surfaces.

For Latency (Y_2), structural lack of fit is non-significant against both reference baselines ($F = 1.20, p = 0.3625$ against center pure error; $F = 1.15, p = 0.3305$ against the saturated additive baseline). The treatment × block interaction is also non-significant ($F = 1.05, p = 0.4906$), so these data did not detect block-related latency variation; this does not establish invariance across all seed blocks, data splits, or hardware conditions. To characterize the functional form of latency growth, five candidate models were compared against empirical measurements, as reported in Table 7. For the log-linear specification, AIC and BIC values are adjusted via the Jacobian of transformation ($+2\sum \ln(y_i) = 1{,}450.8$) to allow direct likelihood comparison on the natural microsecond scale.

Candidate Latency Model	R^2	**Adj** R^2	**AIC** (Adjusted)	**BIC** (Adjusted)	**RMSE** (μs)
Linear (depth)	0.792	0.790	1255.2	1261.1	21.26
Quadratic (depth + depth^2)	0.859	0.857	1202.6	1211.4	17.56
Log-Linear (ln(latency) depth)	0.817	0.815	1218.1	1224.0	19.57
Multi-Factor (depth, depth^2, eta, sub, lam)	0.868	0.863	1199.5	1217.1	17.18
Full CCD Second-Order (All factors + Block)	0.882	0.864	1209.6	1265.5	17.08

Table 7: Comparison of Candidate Regression Models for Inference Latency (Y_2)

The quadratic latency model ($R^2 = 0.859$, adjusted AIC = 1202.6) yields lower AIC than the linear model ($R^2 = 0.792$, AIC = 1255.2) and the log-linear model (adjusted AIC = 1218.1), consistent with the significant x_2^2 term in ANOVA ($F = 39.86, p < 0.0001$). Latency exhibits super-linear growth across the empirical depth range ($d \in \{3, 6, 9\}$), with the quadratic fit capturing accelerating tree traversal costs while avoiding the severe over-prediction of exponential formulations. Comparing standard `predict` against `inplace_predict` across the 140 historical DOE runs shows a mean interface overhead of $40.88 \pm 11.56\,\mu$s (and $32.19 \pm 1.46\,\mu$s in the 50-configuration Revision-v2 interface audit), indicating that `DMatrix` construction and wrapper validation add a consistent baseline latency offset while tree traversal scales super-linearly with depth.

5 Statistical Residual Diagnostics

To evaluate regression assumptions, diagnostic checks were conducted on the second-order Validation RMSE residuals, explicitly distinguishing grouping factors:

(a) **Normality:** Evaluated via Shapiro-Wilk on the 140 second-order residuals: $W = 0.9971, p = 0.9947$, failing to reject the null hypothesis of Gaussian residuals at $\alpha = 0.05$.

(b) **Homoscedasticity Across Seed Blocks vs. Design Coordinates:** Mean-centered Levene's test across the **5 seed blocks** (`block` $1 \ldots 5$) yields $W = 0.1754, p = 0.9507$, confirming uniform residual dispersion across data-split blocks. Median-centered Levene (Brown-Forsythe) testing across the **25 distinct design coordinates** yields $W = 0.6872, p = 0.8552$, because the 5 block observations within each coordinate share a common cell-mean offset ($\bar{y}_{i\cdot} - \hat{y}_i$) and exhibit similar within-coordinate spread around their cell median.

(c) **Reconciliation of Breusch-Pagan with Block and Coordinate Homogeneity:** The Breusch-Pagan test against continuous fitted values $\hat{y}$ provides evidence of heteroscedasticity ($\mathrm{LM} = 70.10, p < 0.0001$). This occurs because Breusch-Pagan regresses squared OLS residuals $e_{ij}^2 = [(\bar{y}_{i\cdot} - \hat{y}_i) + (y_{ij} - \bar{y}_{i\cdot})]^2$ on the design regressors; since deterministic polynomial lack of fit $(\bar{y}_{i\cdot} - \hat{y}_i)^2$ is largest at the high-RMSE $\eta = 0.01$ boundary and compresses in the low-RMSE interior, squared residuals vary systematically across the factor space even though within-cell cross-block variance is homogeneous. HC3 heteroscedasticity-consistent standard errors are used separately for coefficient inference and do not eliminate model misspecification or deterministic polynomial bias in prediction intervals.

(d) **Independence and Run Order:** Durbin-Watson statistic evaluated in actual randomized execution order yields $DW = 1.9918$. The Ljung-Box test ($Q = 4.01, p = 0.5486$) and Wald-Wolfowitz Runs test ($p = 0.8576$) did not detect statistically significant serial dependence in the tested residual sequence. These diagnostics do not establish the absence of all temporal dependence or independently verify the absence of hardware thermal drift.

(e) **Influence and Leverage:** Maximum Cook's distance is $D_{\max} = 0.097$, well below the commonly cited heuristic reference cutoff of 1.0.

Figure 1 displays the 4-in-1 residual diagnostic panel.

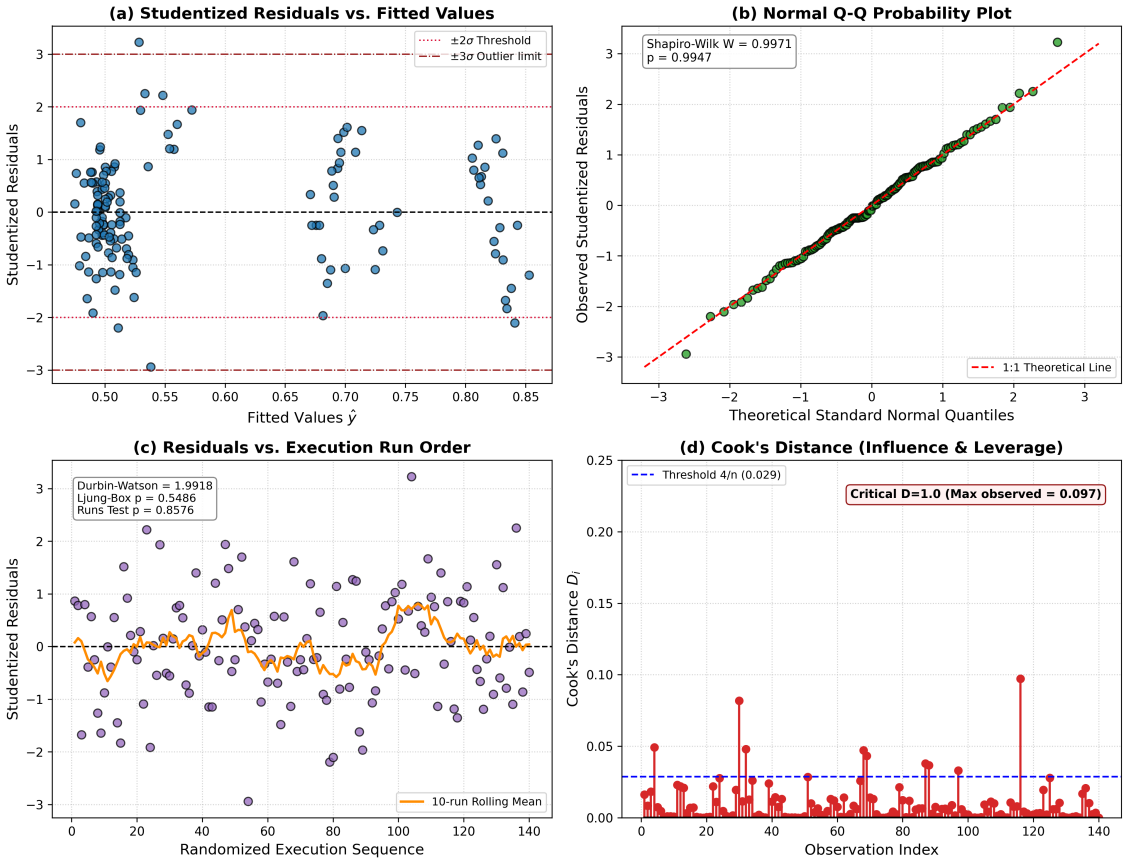


Figure 1: Residual Diagnostics Panel for Second-Order Validation RMSE Model: (a) Studentized residuals vs. fitted values; (b) Normal Q-Q probability plot ($p = 0.9947$); (c) Residuals vs. randomized run execution order ($DW = 1.9918$, Ljung-Box $p = 0.5486$, Runs test $p = 0.8576$); (d) Cook's distance influence plot ($D_{\max} = 0.097$, reference guideline 1.0).

6 Phase 3: Canonical and Ridge Analysis

6.1 Matrix Representation and Stationary Point

The fitted second-order model is expressed in matrix notation:

$$\hat{y} = b_0 + \mathbf{x}^T\mathbf{b} + \mathbf{x}^T\mathbf{B}\mathbf{x} \tag{9}$$

where $b_0 = 0.4994$ (averaging block effects equally), $\mathbf{b} = [-0.1242, -0.0336, -0.0047, -0.0007]^T$, and:

$$\mathbf{B} = \begin{bmatrix} 0.1065 & 0.0169 & -0.0024 & -0.0048 \\ 0.0169 & 0.0261 & -0.0019 & -0.0005 \\ -0.0024 & -0.0019 & 0.0006 & 0.0005 \\ -0.0048 & -0.0005 & 0.0005 & 0.0007 \end{bmatrix} \tag{10}$$

The unconstrained stationary point $\mathbf{x}_0$ satisfies $\mathbf{b} + 2\mathbf{B}\mathbf{x}_0 = \mathbf{0}$:

$$\mathbf{x}_0 = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b} = [0.4171, 1.3524, 15.6944, -6.8009]^T \tag{11}$$

The stationary coordinate along Factor C ($x_3 = 15.6944$) lies far outside the operational cube $[-1, +1]^4$. Its Euclidean distance from the coded origin is $\|\mathbf{x}_0\| = 17.16$, while its Euclidean distance from the boundary of the feasible hypercube is 15.80 coded units. The unconstrained stationary point has a predicted response of $\hat{y}_0 = 0.4161$, an extrapolation artifact indicating that unconstrained quadratic optimization yields an infeasible point.

6.2 Spectral Decomposition and Ridge Analysis

Eigenvalues of $\mathbf{B}$ are $\lambda_1 = 0.000057, \lambda_2 = 0.000873, \lambda_3 = 0.022759, \lambda_4 = 0.110196$ ($\sum \lambda_i = \text{trace}(\mathbf{B}) = 0.133886$). While point estimates are positive, Rademacher wild bootstrap (2,000 replications) reveals that the 95% bootstrap interval for λ_1 spans $[-0.0039, 0.0022]$, with 68.2% of bootstrap samples yielding $\min(\lambda) \leq 0$. The canonical eigenvectors (columns of the orthogonal decomposition matrix $\mathbf{M}$) show that the smallest eigenvalue λ_1 aligns along $\mathbf{v}_1 = [0.023, -0.053, -0.724, 0.687]^T$, representing a contrast between Factor C (subsampling, -0.724) and Factor D (L_2 regularization, $+0.687$), while λ_2 corresponds to $\mathbf{v}_2 = [-0.043, -0.037, -0.687, -0.725]^T$, representing a joint direction along subsampling and regularization. By contrast, the dominant quadratic curvature axes align along Factor B (depth, $\mathbf{v}_3 = [-0.198, 0.978, -0.060, 0.018]^T$, $\lambda_3 = 0.022759$) and Factor A (learning rate, $\mathbf{v}_4 = [-0.979, -0.198, 0.025, 0.044]^T$, $\lambda_4 = 0.110196$). The system is classified as a **stationary or rising ridge**, with curvature along x_3 and x_4 statistically consistent with zero.

Constrained optimization over the hypercube $[-1, 1]^4$ with integer depth $d \in \{3, \ldots, 9\}$ yields the results in Table 8.

Depth	Coded x_1	Coded x_2	Coded x_3	Coded x_4	Optimal η	Predicted RMSE	SE ($\hat{y}$)
3	0.8103	−1.0000	1.00	1.00	0.2173	0.4909	0.0032
4	0.7573	−0.6667	1.00	1.00	0.1985	0.4724	0.0030
5	0.7043	−0.3333	1.00	1.00	0.1814	0.4592	0.0032
6	0.6513	+0.0000	1.00	1.00	0.1658	0.4511	0.0033
7	0.5983	+0.3333	1.00	1.00	0.1515	0.4483	0.0032
8	0.5453	+0.6667	1.00	1.00	0.1384	0.4506	0.0031
9	0.4923	+1.0000	1.00	1.00	0.1265	0.4582	0.0034

Table 8: Best Predicted Validation RMSE at Each Discrete Tree Depth

The constrained single-objective optimum within the hypercube occurs at depth 7 ($\mathbf{x}^*_{\text{SO}}$: $x_1 = 0.5983, x_2 = 0.3333, x_3 = 1.0000, x_4 = 1.0000$) with predicted RMSE 0.4483. Figures 2 and 3 display the response surfaces.

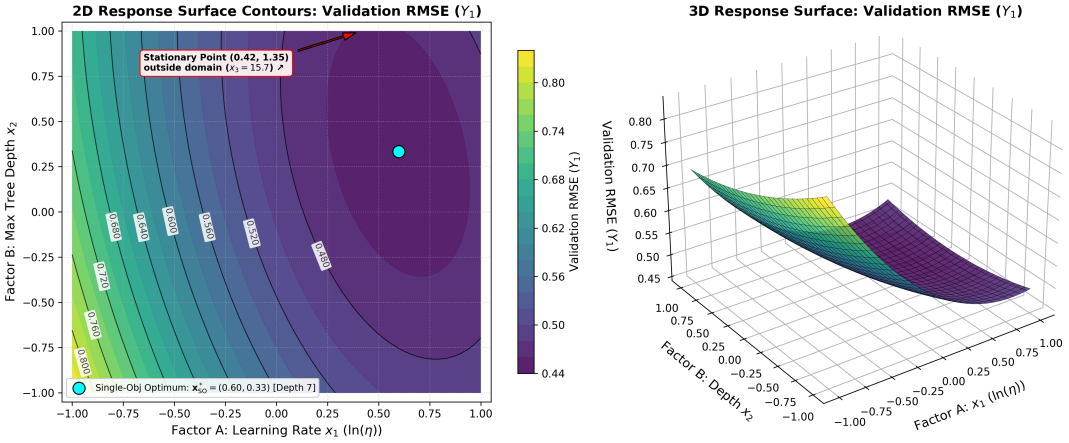


Figure 2: Response Surface Topography for Validation RMSE (Y_1): (a) 2D contour slice across Factor A (ln(η)) and Factor B (depth), passing through the constrained optimum ($x_3 = 1, x_4 = 1$). Arrow indicates trajectory toward unconstrained stationary point outside domain; (b) 3D wireframe projection.

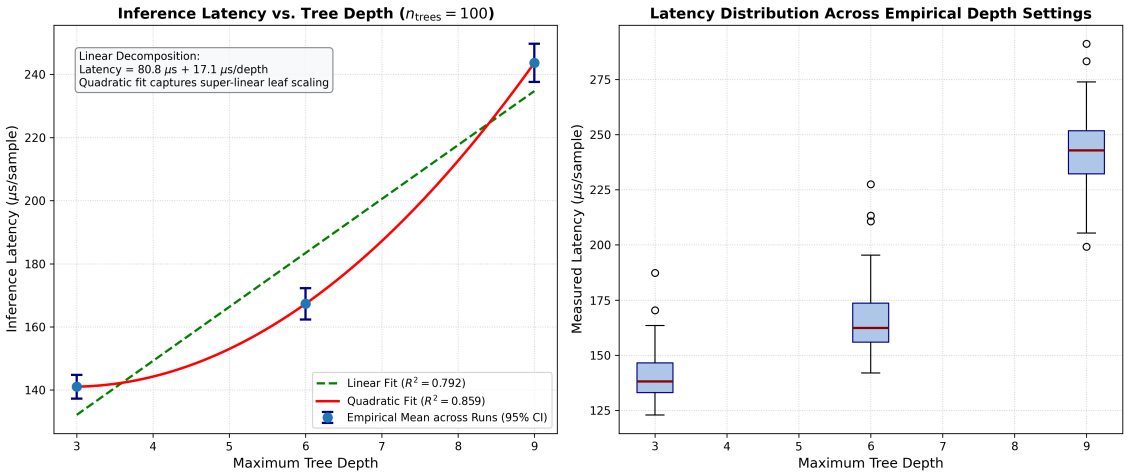


Figure 3: Inference Latency Modelling: (a) Empirical mean latency with 95% CI error bars vs. linear ($R^2 = 0.792$) and quadratic ($R^2 = 0.859$) fits. Note the linear base intercept (≈ 80.8 μs) arising from Python execution overhead; (b) Distribution boxplots across discrete depths.

7 Phase 4: Multi-Objective Desirability Optimization

Minimizing predictive error requires deeper trees, which increases inference latency. To balance predictive error against inference latency, we use Derringer-Suich desirability functions [6]:

$$d_r(Y_r) = \begin{cases} 1.0, & Y_r < L_r \\ \left(\frac{U_r - Y_r}{U_r - L_r}\right)^{s_r}, & L_r \leq Y_r \leq U_r \\ 0.0, & Y_r > U_r \end{cases} \tag{12}$$

The composite desirability D is formulated with explicit importance weights (w_1, w_2):

$$D = (d_1(Y_1)^{w_1} \cdot d_2(Y_2)^{w_2})^{\frac{1}{w_1 + w_2}} \tag{13}$$

In the baseline scenario, equal weights $w_1 = 1, w_2 = 1$ are used with engineering limits $L_1 = 0.450, U_1 = 0.700$ for RMSE and $L_2 = 100.0, U_2 = 180.0$ μs for latency ($s_1 = s_2 = 1.0$).

To select a deployable operating point, desirability D was evaluated across a high-resolution factor grid ($21 \times 7 \times 2 \times 2 = 588$ design coordinates) with discrete depths $d \in \{3, \ldots, 9\}$. While unconstrained maximization pushes toward the upper boundary $\eta \to 0.30$ ($x_1 = 1.0$, reaching $D = 0.6952$) due to the 100-tree ceiling, operating directly on boundary edges risks boundary extrapolation. Holding depth $d = 4$, subsample 1.0, and $\lambda = 0.830$, composite desirability varies smoothly across the interior: $D = 0.6662$ at $\eta = 0.20$, $D = 0.6782$ at $\eta = 0.2324$, $D = 0.6833$ at $\eta = 0.25$, and $D = 0.6943$ at $\eta = 0.30$. We chose the interior coordinate at $x_1 = 0.8500$ ($\eta = 0.2324$) as the selected operational multi-objective compromise configuration $\mathbf{x}^*_{\text{MO}}$:

$$\begin{aligned} \mathbf{x}^*_{\text{MO}} &= [0.8500, -0.6667, 1.0000, -0.0812]^T \\ &\implies (\eta = 0.2324, \text{depth} = 4, \text{subsample} = 1.0000, \lambda = 0.8295) \end{aligned} \tag{14}$$

achieving $D = 0.6782$ ($d_1 = 0.8783, d_2 = 0.5236$). Predicted metrics at $\mathbf{x}^*_{\text{MO}}$ are $\hat{Y}_1 = 0.4804$ and $\hat{Y}_2 = 138.11$ μs.

Table 9 reports sensitivity across alternative specification limits and weights. Across all scenarios, the optimal depth clusters between 3 and 5, indicating that shallow-to-moderate tree depths provide the best balance between accuracy and latency.

Scenario	RMSE Bounds $[L_1, U_1]$	Latency Bounds $[L_2, U_2]$	Weights (w_1, w_2)	Opt Depth	Opt η	Best D
Standard Config	[0.45, 0.70]	[100, 180]	(1, 1)	4	0.3000	0.6952
Strict Latency	[0.45, 0.70]	[90, 140]	(1, 2)	3	0.3000	0.2206
Strict Accuracy	[0.44, 0.60]	[100, 200]	(2, 1)	5	0.2135	0.7293
Narrow Range	[0.46, 0.65]	[110, 160]	(1, 1)	4	0.3000	0.6597

Table 9: Desirability Optimization Sensitivity Analysis Across Bounds and Weights

Figure 4 illustrates the historical DOE development design space, showing individual block run realizations as background observations, configuration-level 5-block means with error bars, the configuration-mean Pareto front, and the operational coordinates $\mathbf{x}^*_{\text{MO}}$ and $\mathbf{x}^*_{\text{SO}}$.

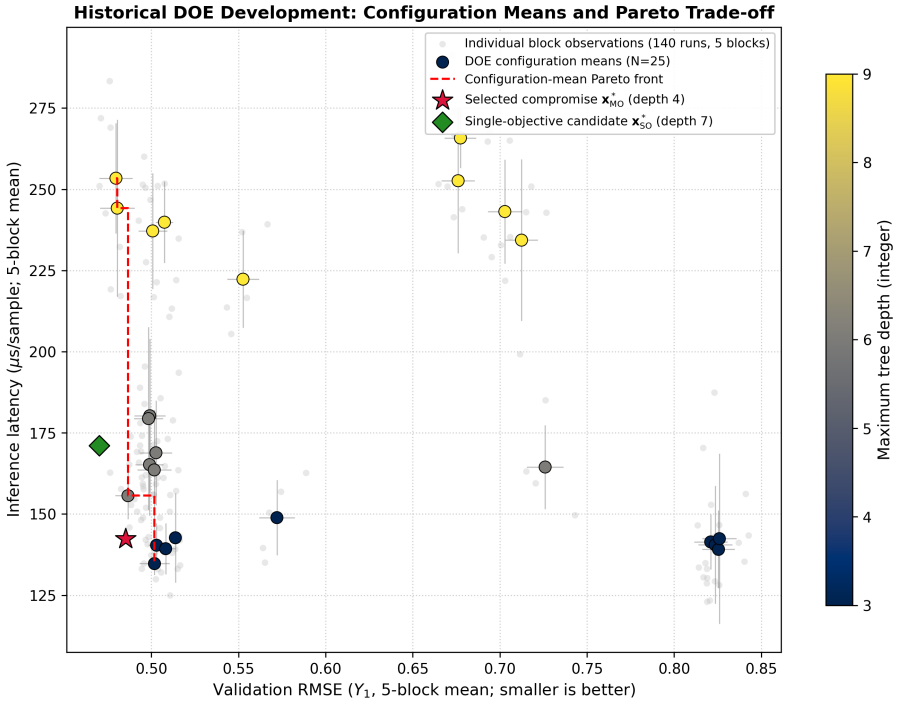


Figure 4: Historical DOE Multi-Objective Trade-Off: Validation RMSE vs. Inference Latency across 5 Seed Blocks. Light gray points show individual block run observations (140 runs across 5 blocks). Filled markers show the 25 unique design configuration means (averaged over blocks) colored by tree depth, with error bars indicating ± 1 SD across blocks. The red dashed line traces the configuration-mean Pareto front. Highlighted operating coordinates include the multi-objective compromise $\mathbf{x}^*_{\text{MO}}$ (depth 4, red star), single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (depth 7, green diamond), and empirical confirmation means ($m = 10$ seeds).

8 Phase 5: Empirical Confirmation and Benchmark Comparison

8.1 Confirmation Experiments at Multi-Objective and Single-Objective Optima

To test surrogate fidelity, $m = 10$ confirmation trials were executed at $\mathbf{x}^*_{\text{MO}}$ across fresh seeds disjoint from the design ($\mathcal{S}_{\text{conf}} = \{505, 606, 707, 808, 909, 1010, 1111, 1212, 1313, 1414\}$). Because fresh seeds introduce block variance (σ^2_{block} estimated with only 4 df) alongside model parameter covariance (MS_E with 121 df), treating block variance uncertainty as negligible would understate interval width. The total theoretical variance for the confirmation mean $\bar{y}_m$ at coordinate $\mathbf{x}_0$ with leverage $h_0 = \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$ is:

$$V_{\text{conf}} = \sigma^2_E \left(\frac{1}{m} + h_0\right) + \sigma^2_{\text{block}} \left(\frac{1}{m} + \frac{1}{5}\right) \tag{15}$$

Under the randomized block design with $a = 28$ runs per block, the expected mean squares satisfy $\mathbb{E}[\text{MS}_E] = \sigma^2_E$ and $\mathbb{E}[\text{MS}_{\text{block}}] = \sigma^2_E + 28\,\sigma^2_{\text{block}}$, giving the unbiased ANOVA estimator $\hat{\sigma}^2_{\text{block}} = (\text{MS}_{\text{block}} - \text{MS}_E)/28$. Substituting $\hat{\sigma}^2_{\text{block}}$ and $\hat{\sigma}^2_E = \text{MS}_E$ into V_{conf} yields the linear combination of independent mean squares required by Satterthwaite's synthesis:

$$\hat{V}_{\text{conf}} = \text{MS}_E \left(\frac{1}{m} + h_0\right) + \frac{\text{MS}_{\text{block}} - \text{MS}_E}{28} \left(\frac{1}{m} + \frac{1}{5}\right) = c_1 \text{MS}_E + c_2 \text{MS}_{\text{block}} \tag{16}$$

where $c_2 = \frac{1/m + 1/5}{28}$ and $c_1 = (1/m + h_0) - c_2$. Propagating variance-component degrees of freedom via Satterthwaite's approximation yields effective degrees of freedom:

$$\nu_{\text{eff}} = \frac{\hat{V}^2_{\text{conf}}}{\frac{(c_1 \text{MS}_E)^2}{121} + \frac{(c_2 \text{MS}_{\text{block}})^2}{4}} \tag{17}$$

For $\mathbf{x}^*_{\text{MO}}$ ($h_0 = 0.1043$), Satterthwaite effective degrees of freedom evaluate to $\nu_{\text{eff}} = 14.0$ ($t_{0.025,14.0} = 2.145$), widening the 95% prediction interval to $[0.4677, 0.4932]$ (compared to $[0.4687, 0.4922]$ under the unadjusted $t_{0.025,121}$ approximation). For the depth-7 single-objective coordinate $\mathbf{x}^*_{\text{SO}}$ ($h_0 = 0.1210$), $\nu_{\text{eff}} = 15.1$ ($t_{0.025,15.1} = 2.131$), producing $[0.4354, 0.4612]$. We note that Satterthwaite prediction intervals pool structural lack of fit (10 df) and treatment $\times$ block interaction (96 df) into MS_E, assuming homogeneous residual variance across design cells without safeguarding against local polynomial bias. Table 10 compares predictions against empirical results.

Configuration / Response Metric	Surrogate Pred ($\hat{y}$)	95% Pred Interval (PI)	Empirical Mean ± SD	Confirmation Status
DOE Multi-Objective Optimum $\mathbf{x}^*_{\mathbf{MO}}$ (Depth 4)				
Validation RMSE (Y_1)	0.4804	$[0.4677, 0.4932]$	0.4853 ± 0.0115	**Pass** (Inside 95% PI)
Holdout Test RMSE	—	—	0.4888 ± 0.0032	Holdout Test Set
Inference Latency (μs)	138.1	$[122.8, 153.4]$	142.3 ± 7.8	**Pass** (Inside 95% PI)
DOE Single-Objective Candidate $\mathbf{x}^*_{\mathbf{SO}}$ (Depth 7)				
Validation RMSE (Y_1)	0.4483	$[0.4354, 0.4612]$	0.4700 ± 0.0093	**Not Confirmed** (Optimism: +0.0217)
Holdout Test RMSE	—	—	0.4691 ± 0.0034	Holdout Test Set
Inference Latency (μs)	187.0	$[171.1, 202.9]$	171.1 ± 10.9	Borderline (At Lower PI)

Table 10: Model Prediction vs. Empirical Confirmation Trials ($m = 10$ Fresh Seeds)

For the multi-objective compromise coordinate $\mathbf{x}^*_{\mathrm{MO}}$ (depth 4), empirical validation RMSE of 0.4853 ± 0.0115 falls directly inside the Satterthwaite prediction interval ($[0.4677, 0.4932]$), providing pointwise confirmation consistent with the surrogate. Empirical latency was $142.33 \pm 7.79\,\mu$s, also falling within $[122.8, 153.4]\,\mu$s. Test RMSE on the external holdout set ($N = 4{,}128$) averaged 0.4888 ± 0.0032. Pointwise confirmation at $\mathbf{x}^*_{\mathrm{MO}}$ does not imply global surrogate accuracy across the entire hypercube.

For the single-objective candidate $\mathbf{x}^*_{\mathrm{SO}}$ (depth 7), the surrogate predicted $\hat{y} = 0.4483$ with Satterthwaite 95% PI $[0.4354, 0.4612]$, while empirical confirmation yielded 0.4700 ± 0.0093. Because empirical validation RMSE lies 0.0088 above the upper bound of the Satterthwaite interval, the prediction was not confirmed. The surrogate exhibited structural optimism of $+0.0217$ ($6.8 \times \mathrm{SE}(\hat{y})$), indicating that quadratic interpolation between sampled design depths $\{3, 6, 9\}$ does not adequately track the diminishing returns of deeper trees. Measured inference latency ($171.1 \pm 10.9\,\mu$s) sits directly on the lower prediction interval bound ($171.1\,\mu$s), representing a borderline confirmation. On the holdout test set, depth 7 achieves an external test RMSE of 0.4691 ± 0.0034, competitive with Bayesian baselines.

8.2 Provenance of Numerical Similarities Across Confirmation and Benchmark Evaluations

Several summary metrics across Table 10 and Table 11 match or nearly match to four decimal places (0.4853, 0.4700, 0.4687, 0.4691, 0.4888) for distinct structural or statistical reasons:

1. **Shared Configuration Hash and Shared 20-Seed Evaluation Pool (`doe-so-historical`, `doe-mo-historical`, and `doe-so-rep-0..19`):** Both the historical benchmark script (`scripts/run_benchmarks.py`) and the Revision-v2 evaluator (`final_evaluation.py`) evaluate frozen configurations across the exact same 20 fresh seeds $\mathcal{S}_{\mathrm{eval}} = \{2001, \ldots, 2020\}$ on the 75/25 split of the 16,512-row development pool and the 4,128-row external holdout test set. Consequently, `doe-so-historical` (`config_sha256 = 0079fbae0ec5b7cb…`: $\eta = 0.1515, d = 7, s = 1.0000, \lambda = 10.0000$) and `doe-mo-historical` (`config_sha256 = b05e8e5cb67e8668…`: $\eta = 0.2324, d = 4, s = 1.0000, \lambda = 0.8295$) yield identical full-precision RMSEs across `results/benchmark.csv` and `final_evaluations.csv` (Test RMSE = 0.469033, Val RMSE = 0.468680 for `doe-so-historical`; Test RMSE = 0.488409, Val RMSE = 0.482071 for

`doe-mo-historical`) while differing in latency due to separate timing sessions. Meanwhile, in Revision-v2, all 20 Repeated DOE Single-Objective replicates select the exact same boundary grid point (`grid_index = 1052`, `config_sha256 = 9d7c4e74ff156870…`: $\eta = 0.1519, d = 7, s = 1.0000, \lambda = 10.0000$) on the predeclared 1,323-point grid. Because `final_evaluation.py` deduplicates model training by `config_sha256` across seeds $2001 \ldots 2020$, all 20 `doe-so-rep-r` records map to the exact same 20 trained XGBoost boosters, yielding identical means (Test RMSE = 0.468591, Val RMSE = 0.468723, rounding to 0.4686 and 0.4687) and zero between-search variance (SD = 0.00000).

2. **Disjoint Seed Sets Agreeing at Four Decimal Places ($\mathcal{S}_{\text{conf}} = \{505, \ldots, 1414\}$ vs. $\mathcal{S}_{\text{eval}} = \{2001, \ldots, 2020\}$):** Phase 5 confirmation evaluates $\mathbf{x}^*_{\text{SO}}$ and $\mathbf{x}^*_{\text{MO}}$ across $m = 10$ seeds completely disjoint from $\mathcal{S}_{\text{eval}}$. At $\mathbf{x}^*_{\text{MO}}$ ($d = 4$), the 10-seed confirmation validation RMSE mean is 0.485286 (SD = 0.0115), whereas the 20-replicate Repeated DOE MO validation RMSE mean across seeds $2001 \ldots 2020$ is 0.485283 (SD = 0.00554); both round to 0.4853 at four decimal places despite using disjoint seed sets and distinct candidate coordinates (10-seed confirmation holdout test RMSE is 0.488774 ± 0.003243, rounding to 0.4888, vs. 0.488409 ± 0.003191, rounding to 0.4884, on the 20 benchmark seeds and 0.48997 or 0.4900 at 4 decimal places for Repeated DOE MO). Likewise, at $\mathbf{x}^*_{\text{SO}}$ ($d = 7$), the 10-seed confirmation validation RMSE mean is 0.470000 (SD = 0.0093) and the 20-replicate Constrained TPE validation RMSE mean on seeds $2001 \ldots 2020$ is 0.469962, both rounding to 0.4700 at four decimal places; on the holdout test set, the 10-seed confirmation mean at $\mathbf{x}^*_{\text{SO}}$ is 0.469124 (SD = 0.003437, rounding to 0.4691) versus 0.469033 (SD = 0.002585, rounding to 0.4690, differing by 91.25×10^{-6} RMSE) on the 20 benchmark seeds.

8.3 Benchmark Evaluation Against Random Search and Optuna TPE

To compare the pre-planned Design of Experiments framework against modern heuristic and Bayesian search algorithms under equal evaluation budgets (140 function evaluations per search), we executed a prospective benchmark campaign across $N = 20$ independent search replicates per optimizer:

1. **Unguided Random Search:** 20 independent replicates of 140 evaluations uniformly sampled across the 4-factor domain.

2. **Single-Objective TPE:** 20 independent replicates of 140 evaluations optimizing validation RMSE via Tree-structured Parzen Estimators [4, 5].

3. **Constrained TPE:** Single-objective TPE with an operational inference latency constraint ($\leq 145\,\mu$s) evaluated via single-threaded single-sample prediction latency measured online during trial sampling (30 calls without warmup).

4. **Multi-Objective TPE:** Evaluates validation RMSE and online 30-call single-sample prediction latency simultaneously to construct a candidate Pareto front, followed by post-hoc operating coordinate selection via Derringer-Suich desirability.

5. **Repeated Pre-Planned DOE Formulations:** 20 independent replicates of the pre-planned 140-run Face-Centered Central Composite Design across fresh 5-block seed quintuplets (seeds 3001–3100), evaluating both the multi-objective compromise $\mathbf{x}^*_{\text{MO}}$ (depth 4) and the single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (depth 7).

Across the six methods, winning hyperparameter configurations from the 20 search replicates (122 selection records representing 95 distinct configuration hashes, yielding 2,440 final evaluation rows) were serialized into an immutable manifest (`finalized_selections.json`, verified by SHA-256) prior to holdout evaluation. Across the prospective benchmark campaign, 17,077

XGBoost model fits and 1,640,810 timed prediction evaluations (1,842,080 total prediction calls including 201,270 untimed warmup calls) were executed.

Crucially, we clarify the history of the holdout test set: the historical v1.0.0 experiments recorded test results during development, whereas the revised Revision-v2 pipeline programmatically prevented access to those test labels during all search iterations, surrogate fitting, and desirability calculations until winning configurations were frozen in `finalized_selections.json`. While these safeguards enforce pipeline isolation during the revised experiments, they cannot undo earlier historical exposure of the same dataset. Each frozen hyperparameter configuration was then retrained from scratch and evaluated across 20 fresh seeds (seeds 2001–2020, $\mathcal{S}_{\text{eval}}$) on the external holdout test set under dedicated single-threaded latency profiling (2×500 timed single-row calls per seed with 50 warmup calls per session, pinned to CPU Core 0). Because both historical v1.0.0 scripts (`scripts/run_confirmation.py` and `scripts/run_benchmarks.py`) also applied Win32 CPU core 0 affinity pinning, the latency difference between historical single-run snapshots and the prospective campaign reflects between-session system timing drift (SD = 1.28 μs across repeated sessions) rather than unpinned execution.

Table 11 summarizes the benchmark comparison, reporting Panel A (the prospective 20-replicate empirical evidence with between-search standard deviations, retraining standard deviations, and benchmark constraint feasibility) alongside Panel B (the historical v1.0.0 baseline snapshot).

Optimization Method	**Search Basis** (N Reps)	**Val RMSE** Mean ± SD	**Holdout Test RMSE** Mean ± SD	**Retrain** $\boldsymbol{\sigma}_{\text{eval}}$	**Latency** (μs ± SD)	**Feasible** ($\leq 145\,\mu$s)
Panel A: Prospective Revision-v2 Full Benchmark ($N = 20$ Search Replicates)						
Repeated DOE MO ($\mathbf{x}^*_{\text{MO}}$)	RSM (N=20)	0.4853 ± 0.0055	0.4900 ± 0.0073	0.0033	122.3 ± 2.7	20/20
Multi-Objective TPE	Parzen (N=20)	0.4881 ± 0.0093	0.4926 ± 0.0108	0.0036	121.5 ± 3.8	20/20
Constrained TPE ($\leq 145\,\mu$s)	Parzen (N=20)	0.4700 ± 0.0022	0.4704 ± 0.0032	0.0034	144.3 ± 7.1	9/20
Repeated DOE SO (d=7)	RSM (N=20)	0.4687 ± 0.0000	0.4686 ± 0.0000	0.0030	150.7 ± 0.7	0/20
Single-Objective TPE	Parzen (N=20)	0.4675 ± 0.0008	0.4669 ± 0.0012	0.0031	188.9 ± 12.1	0/20
Unguided Random Search	Uniform (N=20)	0.4687 ± 0.0014	0.4695 ± 0.0021	0.0030	181.1 ± 19.0	0/20
Panel B: Historical Baseline Snapshot (v1.0.0, Single Search Replicate)						
Historical DOE MO ($\mathbf{x}^*_{\text{MO}}$)	RSM (Single)	0.4821	0.4884	—	119.8	Yes
Historical Multi-Obj TPE	Parzen (Single)	0.4812	0.4870	—	118.8	Yes
Historical Constrained TPE	Parzen (Single)	0.4783	0.4823	—	125.4	Yes
Historical DOE SO (d=7)	RSM (Single)	0.4687	0.4690	—	147.8	No
Historical Bayesian TPE	Parzen (Single)	0.4735	0.4726	—	193.2	No
Historical Random Search	Uniform (Single)	0.4696	0.4707	—	198.1	No

Table 11: Empirical Benchmark Comparison across 20 Fresh Seeds on Fixed Holdout Test Set. Panel A reports prospective Revision-v2 evidence across $N = 20$ independent search replicates (140 fits per replicate) evaluated across 20 retraining seeds $\mathcal{S}_{\text{eval}}$ on the external holdout set ($N = 4,128$). Panel B preserves the historical v1.0.0 single-replicate baseline snapshot for reference.

Table 12 details the empirical Pareto hypervolume comparisons across candidate search frontiers and the holdout evaluated selections. Importantly, rows in Panel A and Panel B represent distinct evaluation protocols and candidate pools rather than a single homogeneous two-sample experiment: MO-TPE candidate fronts are evaluated across 20 searches of 140 trials

on development split 42 under 30-call timing; Repeated DOE candidate fronts average 25 FCCD coordinates across 5 block seeds per replicate in 3001 . . . 3100 under 30-call timing; the Fixed Full DOE Evaluated Front evaluates a single deterministic 27-point set (25 FCCD points plus $\mathbf{x}^*_{\mathrm{MO}}$ and $\mathbf{x}^*_{\mathrm{SO}}$) once on split 42 under 1,000-call warmed timing; and Panel B evaluates the 122 frozen selection records (95 distinct configurations) on the external holdout test set across 20 seeds and 2 timing sessions. Consequently, the Split-42 difference row (-1.5249 at $[0.60, 250.0]$; -1.9105 at $[0.65, 275.0]$) is a descriptive one-sample comparison of the 20 stochastic MO-TPE fronts against the fixed 27-point DOE reference on split 42, not a two-sample test across independent data splits. Likewise, in the historical v1.0.0 snapshot (`results/benchmark_summary.json`), the reported 16.50 vs. 14.82 (+11.4%) hypervolume compared the full 25-point DOE design against a single selected MO-TPE point; when restricted to single deployed points ($\mathbf{x}^*_{\mathrm{MO}}$ vs. MO-TPE), historical hypervolume is 14.53 vs. 14.82 ($+ - 1.9\%$).

Frontier / Evaluation Protocol	**Candidate Pool (N)**	**Non-Dom. Points**	**HV at** $[0.60, 250]$ (Mean ± SD)	**HV at** $[0.65, 275]$ (Mean ± SD)
Panel A: Development Candidate Frontiers (Distinct Evaluation Protocols Noted per Row)				
Multi-Objective TPE Candidates (Single Split 42, 30-Call Search)	140×20	9.35 ± 2.3	18.3834 ± 0.3301	30.0535 ± 0.4123
Repeated DOE Candidate Fronts (5-Block Means Across Replicates)	25×20	6.10 ± 1.4	15.6128 ± 0.7268	26.6847 ± 0.8786
Fixed Full DOE Evaluated Front (Single Split 42, 27 Configs)	27	4	16.8585	28.1430
Historical DOE 2-Point Set (Single Split 42, $\mathbf{x}^*_{\mathrm{MO}}$ & $\mathbf{x}^*_{\mathrm{SO}}$)	2	2	16.0363	26.9168
Difference: Fixed Full DOE Front − MO-TPE Mean (Split 42)	—	—	-1.5249 (1-sample $p < 0.0001$)	-1.9105 (1-sample $p < 0.0001$)
Panel B: Holdout Non-Dominated Set (122 Records, 95 Configs, 20 Seeds)				
Non-Dominated Set (12 Distinct Configs)	122	12	17.6714	28.9900
Holdout Frontier Seed-Bootstrap ($B = 2{,}000$ Resamples over 20 Seeds)	122	13.42 ± 1.46	17.6923 ± 0.0564	29.0170 ± 0.0688
⌞ Repeated DOE MO Selections	20	3 (100.0%)	—	—
⌞ Multi-Objective TPE Selections	20	3 (100.0%)	—	—
⌞ Constrained TPE Selections	20	5 (100.0%)	—	—
⌞ Single-Objective TPE Selections	20	1 (91.5%)	—	—

Table 12: Pareto Hypervolume Analysis Across Development Frontiers and External Holdout Evaluated Selections. Panel A summarizes development candidate fronts across their respective evaluation protocols (MO-TPE candidates on single development split 42 under 30-call timing, Repeated DOE candidates averaged across five seed blocks per replicate, and the fixed 27-point DOE evaluated frontier on development split 42). Panel B reports the non-dominated set among the 122 frozen evaluated selection records (95 distinct configurations) across 20 retraining seeds on the external holdout test set alongside paired cluster bootstrap uncertainty ($B = 2,000$ resamples over the 20 evaluation seeds).

Figure 5 contrasts development candidate frontiers on development split 42 (left) with genuine holdout test set evaluations of frozen selections across 20 retraining seeds (right).

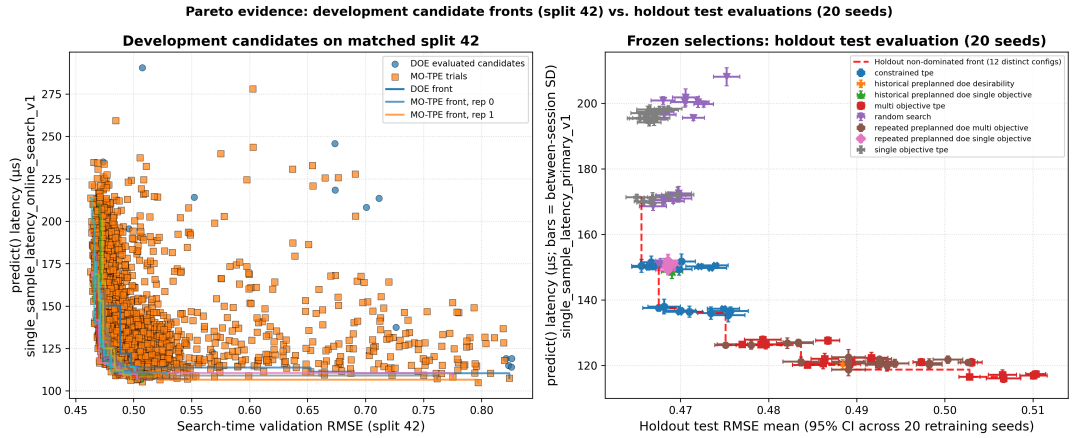


Figure 5: Pareto Evidence Architecture: (Left) Development candidate Pareto fronts on matched split 42 under the online 30-call latency protocol, showing evaluated DOE points, MO-TPE search trial clouds, and replicate-level candidate Pareto steps; (Right) Evaluated frozen selections on the external holdout test set across 20 retraining seeds, with horizontal error bars showing 95% confidence intervals for holdout test RMSE and vertical error bars showing between-session latency standard deviations. The red dashed line traces the empirical holdout non-dominated front comprising 12 distinct configurations.

Figure 6 contrasts optimization sample efficiency across the prospective 20-replicate search trajectories (Panel a) and the historical v1.0.0 single-campaign trace (Panel b).

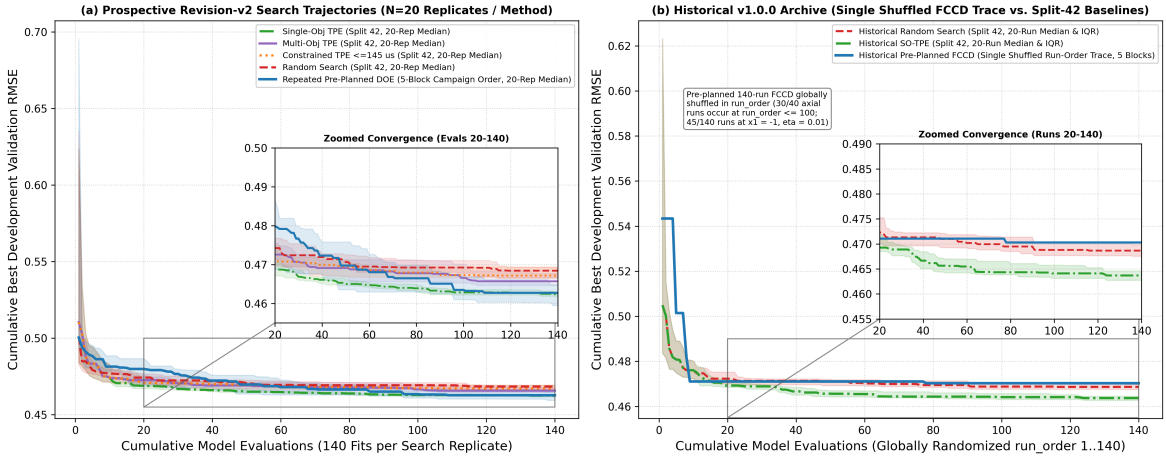


Figure 6: Sample Efficiency and Cumulative Best Validation RMSE Across 140 Evaluations: (a) Prospective Revision-v2 20-replicate trajectories (`optimizer_trials.csv` and `doe_selection_runs.csv`, median ± IQR shaded bands), showing Single-Objective TPE, Constrained TPE, Multi-Objective TPE, and Random Search on split 42 alongside Repeated Blocked DOE (140 pre-planned FCCD runs across 5 blocks per replicate, plotted both as raw single-block run minimums and 5-block configuration means across the 25 distinct coordinates); (b) Historical v1.0.0 single-campaign trace (`results/runs.csv`, where the 140 pre-planned runs were executed in a single globally shuffled run order across blocks so that 30 of 40 axial runs occurred before run 100) compared against the single Split-42 baseline trajectories.

The prospective empirical benchmarks establish six key scientific findings:

1. **Single-Objective Predictive Performance vs. Latency:** Single-Objective TPE achieves the numerically lowest observed mean Test RMSE (0.46691 ± 0.00116), selecting

deep trees ($d \in \{7, 8, 9\}$) with an average inference latency of $188.93 \pm 12.07\,\mu$s (inplace $157.14\,\mu$s). Neither Single-Objective TPE nor Repeated DOE Single-Objective satisfies the $\leq 145\,\mu$s latency constraint (0/20 feasible). Repeated DOE Single-Objective ($\mathbf{x}^*_{\text{SO}}$, depth 7) achieves a mean Test RMSE of 0.46859 at $150.74 \pm 0.72\,\mu$s latency (20.2% lower latency than SO-TPE).

2. **What Repeated DOE Replicates Represent and Separation of Selection vs. Retraining Variance:** Each of the $N = 20$ Repeated DOE replicates ($r \in \{0, \ldots, 19\}$) executes a fresh 140-run blocked FCCD across 5 disjoint block seeds ($3001 + 5r$ to $3005 + 5r$, spanning seeds 3001 . . . 3100) drawn from the 16,512-row development pool, fits OLS second-order surfaces for validation RMSE and latency, and selects candidates over a deterministic 1,323-point predeclared grid (`doe_selection_protocol.json`: 21 coded learning-rate levels $x_1 \in [-1, +1] \times 7$ integer depths $d \in \{3, \ldots, 9\} \times 3$ coded subsample levels $x_3 \in \{-1, 0, +1\} \times 3$ coded regularization levels $x_4 \in \{-1, 0, +1\}$). Because the single-objective minimum on this grid consistently lies at the same boundary coordinate (`grid_index = 1052`, $x = [+0.6, +1/3, +1.0, +1.0]^T \implies \eta = 0.1519, d = 7, s = 1.0, \lambda = 10.0000$), all 20 DOE-SO replicates select the identical configuration hash (`9d7c4e74ff156870…`), yielding between-search selection SD = 0.00000; its non-zero uncertainty is captured entirely by the conditional retraining standard deviation $\sigma_{\text{eval}} = 0.00304$ across the 20 evaluation seeds 2001 . . . 2020. For DOE-MO, desirability optimization over feasible grid points selects across 12 distinct grid configurations in $d \in \{4, 5\}$, yielding between-search SD = 0.00732 alongside conditional retraining $\sigma_{\text{eval}} = 0.00330$.

3. **Multi-Objective Statistical Comparison (Primary Unpaired Welch vs. Secondary Index-Paired Test):** Comparing multi-objective solutions, Repeated DOE ($\mathbf{x}^*_{\text{MO}}$, depth 4) achieves a mean Test RMSE of 0.48997 ± 0.00732 at $122.29 \pm 2.70\,\mu$s latency (inplace $89.73\,\mu$s), closely matching Multi-Objective TPE which achieves 0.49255 ± 0.01080 at $121.46 \pm 3.75\,\mu$s (inplace $88.84\,\mu$s). Because replicate index $r \in \{0, \ldots, 19\}$ pairs DOE block seeds $3001 + 5r \ldots 3005 + 5r$ with Optuna sampler seed $1000 + r$ on split 42 rather than a shared random block, Welch's two-sample t-test is the primary inferential comparison ($\Delta = -0.00258$ RMSE, $t = -0.8849, p = 0.3826$), while the replicate-index paired t-test ($t = -0.966, p = 0.3463$) is reported as a secondary sensitivity check. Because no pre-registered equivalence margin was specified, this non-significant difference indicates no detected performance gap rather than formal statistical equivalence. Both multi-objective methods achieve 100% benchmark constraint feasibility (20/20).

4. **Development Hypervolume vs. Holdout Non-Dominated Set and Seed-Bootstrap Uncertainty:** On development split 42 under 30-call search timing, MO-TPE candidate fronts attain a mean hypervolume of 18.3834 ± 0.3301 at reference $[0.60, 250.0]$ (30.0535 ± 0.4123 at $[0.65, 275.0]$), exceeding the fixed 27-point DOE candidate frontier (16.8585, 1-sample difference $-1.5249, p < 0.0001$; at $[0.65, 275.0]$, 28.1430, difference $-1.9105, p < 0.0001$), while Repeated DOE 5-block means achieve 15.6128 ± 0.7268. On the external holdout test set across 20 retraining seeds, 12 distinct configurations form the empirical non-dominated set among the 122 frozen selections (hypervolume 17.6714 at $[0.60, 250.0]$ and 28.9900 at $[0.65, 275.0]$): 3 Repeated DOE MO, 3 MO-TPE, 5 Constrained TPE, and 1 SO-TPE. Paired cluster bootstrap resampling ($B = 2{,}000$ resamples over the 20 shared evaluation seeds 2001 . . . 2020) yields a holdout frontier hypervolume of 17.6923 ± 0.0564 (95% CI $[17.5848, 17.8061]$) at $[0.60, 250.0]$ and 29.0170 ± 0.0688 (95% CI $[28.8861, 29.1579]$) at $[0.65, 275.0]$ (joint seed-plus-session bootstrap: 17.6890 ± 0.0727, 95% CI $[17.5507, 17.8420]$). Across the 2,000 seed-bootstrap frontiers, Repeated DOE MO, MO-TPE, and Constrained TPE each contribute at least one non-dominated configuration in 100.0%, 100.0%, and 100.0% of resamples (mean frontier points: 2.76, 3.16, and 6.03), while SO-TPE contributes in 91.5% (mean 1.38 points).

5. **Constrained TPE Feasibility Dynamics:** Constrained TPE achieves 20/20 feasibility during search under online 30-call timing, but yields 9/20 (45%) feasibility under the 1,000-call benchmark evaluation protocol ($136.70 \pm 0.82\,\mu$s for depth 6 selections, 9/9 feasible; $150.45 \pm 0.67\,\mu$s for depth 7 selections, 0/11 feasible). Plausible explanations include noisy 30-call search measurements near the boundary, protocol differences (30 calls without warmup vs. 1,000 calls with warmup), and winner's-curse selection bias near the $145\,\mu$s threshold.

6. **Computational Budget Reconciliation and Wall-Clock Cost:** Table 13 reconciles all model fits, distinct coordinates, timed inferences, untimed warmup calls, and per-replicate search wall-clock times across both campaigns. Although each search replicate executes exactly 140 XGBoost fits, mean training time per fit and wall-clock time per replicate vary systematically with the depth and learning-rate distribution sampled by each algorithm: Repeated Blocked DOE averages 36.49 ± 0.19 s per replicate (238.6 ms/fit, shared across DOE-MO and DOE-SO), Multi-Objective TPE averages 26.24 ± 1.80 s (174.2 ms/fit), Constrained TPE averages 26.50 ± 2.70 s (176.9 ms/fit), Random Search averages 32.61 ± 1.63 s (221.9 ms/fit), and Single-Objective TPE requires 54.03 ± 3.05 s (370.2 ms/fit) because adaptive concentration in depths 7–9 increases tree construction cost.

Phase / Method Component	**Search Reps**	**XGBoost Fits**	**Distinct Coords**	**Timed Inferences**	**Warmup Calls**	**Wall Time / Rep** (s, Mean ± SD)
Panel A: Historical Single-Campaign Baseline ($v1.0.0$, 160 DOE + Confirmation Fits)						
Phase 1: 2^4 Factorial + 4 Center (5 Blocks)	1	100	17	100,000	10,000	—
Phase 2: Face-Centered Axial (5 Blocks)	1	40	8	40,000	4,000	—
Phase 5: Confirmation ($\mathbf{x}^*_{\mathrm{MO}}$ & $\mathbf{x}^*_{\mathrm{SO}}$, $m = 10$)	2	20	2	20,000	2,000	—
Panel B: Prospective Revision-v2 Campaign (Total = 17,077 Fits, 1,640,810 Timed Inferences)						
Repeated Pre-Planned DOE (MO + SO Shared)	20	2,800	25/rep	84,000	28,000	36.49 ± 0.19
Multi-Objective TPE (140 Trials/Rep)	20	2,800	140/rep	84,000	28,000	26.24 ± 1.80
Constrained TPE (140 Trials/Rep)	20	2,800	140/rep	84,000	28,000	26.50 ± 2.70
Single-Objective TPE (140 Trials/Rep)	20	2,800	140/rep	84,000	28,000	54.03 ± 3.05
Random Search (140 Trials/Rep)	20	2,800	140/rep	84,000	28,000	32.61 ± 1.63
Split-42 Full DOE Baseline (25 + 2 Opt)	1	27	27	810	270	—
Primary Latency Sessions (5 × 122 Records)	5	610	95	1,220,000	61,000	—
Holdout Retraining (122 Records ×20 Seeds)	—	2,440	95	0	0	—
Prospective Campaign Total	—	**17,077**	—	**1,640,810**	**201,270**	**1,842,080** calls

Table 13: Complete Computational Budget Ledger and Wall-Clock Execution Summary across the Historical v1.0.0 Baseline and the Prospective Revision-v2 Campaign (17,077 Fits, 1,640,810 Timed Inferences, 201,270 Warmup Calls, and 1,842,080 Total Single-Row Calls).

9 Discussion and Limitations

The empirical findings clarify the trade-offs between classical Design of Experiments and adaptive Bayesian optimization:

1. **Parametric Attribution and Variance Decomposition vs. Adaptive Coordinate Exploration:** Pre-planned blocked DOE provides orthogonal main-effect and interaction attribution, formal non-planarity testing ($F = 105432.31$ against center pure error; $F = 7{,}342.69$ across the 5 seed blocks), multi-scale lack-of-fit diagnostics ($F = 62.21$), and high selection stability across replicates (between-search SD = 0.00000 for DOE-SO and 0.00732 for DOE-MO). However, spending 140 runs to replicate 25 fixed coordinates across 5 blocks limits spatial coverage and allocates 45/140 runs (32.14%) to the severe underfitting boundary ($\eta = 0.01$). Conversely, adaptive TPE explores 140 distinct coordinates and concentrates trials near promising interior regions, achieving lower single-objective Test RMSE

(0.46691 vs. 0.46859) and higher single-split development candidate hypervolume (18.3834 vs. 16.8585), at the expense of higher selection variance and no parametric decomposition.

2. **Stochastic Seed Blocking:** Random complete block design isolated an estimated $\sigma_{\text{block}} \approx 0.0077$ RMSE in unexplained residual variance (ICC = 0.4069, or 40.7%; REML ICC = 0.4069). While primary factors account for $> 90\%$ of total sum of squares across the wide hypercube, this seed-to-seed partition shift exceeds the differences separating competitive configurations near the optimum, demonstrating the importance of blocking when comparing configurations across data splits.

3. **Structural Lack of Fit and Surrogate Optimism:** Second-order polynomials cannot globally match the asymmetric saturation of gradient boosting loss across an order-of-magnitude learning-rate range. This structural limitation produced significant lack of fit ($F = 62.21, p = 7.78e-41$, $\sqrt{\text{MS}_{\text{LoF}}} = 0.0297$) and surrogate optimism at intermediate depth 7 (bias = +0.0217, $6.8 \times \text{SE}(\hat{y})$). Because Satterthwaite prediction intervals pool structural lack of fit (10 df) and treatment $\times$ block interaction (96 df) into MS_E, they widen intervals under unexplained variance but do not protect against localized deterministic polynomial bias.

4. **Latency Protocol Sensitivity and Interface Overhead:** Single-sample inference latency follows super-linear growth across discrete depths ($R^2 = 0.859$), while standard `predict` incurs a consistent wrapper and `DMatrix` construction overhead of $32.19 \pm 1.46\,\mu$s ($40.88 \pm 11.56\,\mu$s in historical runs) over `inplace_predict`. Furthermore, Constrained TPE's drop from 20/20 search feasibility (30 un-warmed calls) to 9/20 benchmark feasibility (1,000 warmed calls) demonstrates that online latency constraints require warmed multi-call verification and safety margins near operational thresholds.

5. **Scope and Dimensionality Scaling:** These conclusions apply to XGBoost with 100 boosting trees on the California Housing dataset across $k = 4$ hyperparameters. Because full factorial and central composite designs scale as $2^k + 2k + n_c$ per block, pre-planned second-order designs become prohibitive in higher dimensions ($k > 8$) without fractional factorial screening (Resolution IV/V) or hybrid screening-plus-Bayesian refinement.

10 Code Availability

The code and research materials are publicly available at: https://github.com/ChristosHussein/xgboost-doe-rsm

11 Conclusion

This study evaluated a pre-planned blocked Design of Experiments (DOE) and Response Surface Methodology (RSM) framework for multi-objective hyperparameter optimization in gradient boosted decision trees. Using a randomized complete block design across 5 data-split blocks ($N = 140$ runs across 25 FCCD coordinates), the framework isolated stochastic partition variance ($\sigma_{\text{block}} \approx 0.0077$, ICC = 0.4069), confirmed strong non-planarity across six error denominators ($F = 105432.31$ against center pure error; $F = 7{,}342.69$ across blocks), and identified an interior multi-objective compromise configuration ($\mathbf{x}^*_{\text{MO}}$, depth 4) via Derringer-Suich desirability. Empirical confirmation across $m = 10$ fresh seeds validated the Satterthwaite prediction intervals at $\mathbf{x}^*_{\text{MO}}$ (0.4853 ± 0.0115 validation RMSE; $142.33 \pm 7.79\,\mu$s latency), while exposing structural polynomial optimism at the single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (+0.0217 RMSE bias at depth 7).

Prospective benchmark evaluations comprising 17,077 model fits across 20 independent search replicates and 20 fresh retraining seeds on an external holdout test set demonstrated the complementary strengths of pre-planned DOE and adaptive Bayesian optimization. Single-Objective TPE achieved the lowest mean holdout Test RMSE (0.46691 ± 0.00116) at higher latency ($188.93\,\mu$s) and longer search wall-clock time (54.03 s), whereas Repeated DOE Single-Objective achieved deterministic grid selection (0.46859, between-search SD = 0.00000, retraining $\sigma_{\text{eval}} = 0.00304$) with 20.2% lower latency ($150.74\,\mu$s). In the latency-constrained multi-objective regime ($\leq 145\,\mu$s), Repeated DOE ($\mathbf{x}^*_{\text{MO}}$) and Multi-Objective TPE achieved 100% benchmark feasibility (20/20) with no statistically significant difference in holdout Test RMSE (0.48997 vs. 0.49255, Welch $p = 0.38$) at comparable single-sample latency ($122.3\,\mu$s vs. $121.5\,\mu$s). While MO-TPE achieved higher development candidate hypervolume on split 42 (18.3834 vs. 16.8585 for the 27-point DOE front), holdout evaluation across 20 retraining seeds yielded a 12-configuration empirical non-dominated set (HV = 17.6714, seed-bootstrap 95% CI $[17.5848, 17.8061]$) populated jointly by DOE and TPE selections. Classical blocked DOE and RSM thus provide a transparent, variance-shielded complement to adaptive Bayesian search for latency-constrained tabular machine learning.

References

[1] D. C. Montgomery, *Design and Analysis of Experiments*, 9th ed. Hoboken, NJ: John Wiley & Sons, 2017, ch. 4 (Randomized Complete Block Designs), ch. 5 (Introduction to Factorial Designs), ch. 6 (2^k Factorial Designs), ch. 7 (Blocking and Confounding in the 2^k Factorial Design), ch. 10 (Fitting Regression Models), ch. 11 (Response Surface Methods and Designs), and ch. 13 (Experiments with Random Factors).

[2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[3] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012.

[4] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in *Advances in Neural Information Processing Systems*, vol. 24, 2011, pp. 2546–2554.

[5] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.

[6] G. Derringer and R. Suich, "Simultaneous optimization of several response variables," *Journal of Quality Technology*, vol. 12, no. 4, pp. 214–219, 1980.

[7] G. E. P. Box and K. B. Wilson, "On the experimental attainment of optimum conditions," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 13, no. 1, pp. 1–45, 1951.

[8] R. H. Myers, D. C. Montgomery, and C. M. Anderson-Cook, *Response Surface Methodology: Process and Product Optimization Using Designed Experiments*, 4th ed. Hoboken, NJ: John Wiley & Sons, 2016, ch. 6 (Analysis of Second-Order Response Surfaces) and ch. 7 (Multiple Response Optimization).